

Inter-City Freight Load Matching Marketplace

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for the Inter-City Freight Load Matching Marketplace — a B2B freight brokerage platform that connects commercial shippers with trucking carriers. Shippers post cargo loads, carriers submit competitive bids, and the system automates milestone-based payment release upon verified proof-of-delivery (POD) sign-off.

1.2 Scope

The system encompasses load posting, bid management, carrier–shipper matching, shipment tracking, proof-of-delivery verification, and escrow-based payment processing. It serves as a digital exchange that reduces brokerage overhead, improves freight utilization, and ensures payment security for all parties.

1.3 Definitions and Acronyms

Acronym / Term	Definition
Shipper	A commercial entity that needs to transport cargo between cities
Freight Carrier	A trucking company or independent operator that transports cargo
POD	Proof of Delivery — documented evidence that cargo was delivered
Escrow	A financial arrangement where payment is held by the system until delivery
Load	A cargo consignment posted by a shipper for transport
Bid	A price quotation submitted by a carrier for a specific load

2. Requirements Table

2.1 Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a registered Shipper to post a new cargo load by specifying origin city, destination city, cargo type, weight (in kg), dimensions, pickup date, and delivery deadline.	High	Pass: A load record is created and visible on the marketplace within 5 seconds of submission with all mandatory fields populated. Fail: The load is not created, or any mandatory field is missing.	Shippers must be able to list their freight requirements so that carriers can discover and bid on available loads.
FR-002	Functional	The system shall allow a registered Freight Carrier to search available loads by filtering on origin city, destination city, cargo type, and date range, and to submit a monetary bid on any open load.	High	Pass: Search returns all matching loads within 3 seconds; a bid is recorded against the selected load with the carrier's quoted price and estimated transit time. Fail: Search returns incorrect results, or bid submission fails.	Carriers need to find relevant loads matching their route capacity and submit competitive pricing to win contracts.
FR-003	Functional	The system shall enable the Shipper to review all bids received on a posted load, compare carrier ratings, and accept exactly one bid — thereby awarding the shipment contract to the	High	Pass: Upon acceptance, the load status changes to “Awarded,” the winning carrier receives a confirmation notification, and all other bidders receive a rejection notification. Fail: Multiple bids are accepted for the	A structured bid evaluation process ensures shippers select the most suitable carrier based on price, rating, and transit time.

		selected carrier.		same load, or notifications are not sent.	
FR-004	Functional	The system shall allow the Freight Carrier to upload a digitally signed Proof-of-Delivery (POD) document — including recipient name, signature image, delivery timestamp, and optional photo evidence — once the shipment reaches the destination.	High	Pass: The POD is stored, the shipment status updates to “Delivered — Pending Verification,” and the shipper receives a notification to verify delivery. Fail: POD upload fails, or the shipment status does not update.	A verifiable POD is the contractual trigger for releasing escrowed funds and protects both parties against disputes.
FR-005	Functional	The system shall automatically release the escrowed milestone payment to the Freight Carrier's registered bank account within 24 hours after the Shipper confirms the POD or after a 72-hour auto-approval window elapses without shipper objection.	Medium	Pass: Payment is transferred to the carrier's account within the specified time frame, and both parties receive a payment confirmation receipt. Fail: Payment is not released after confirmation/auto-approval, or payment is released before POD verification.	Automated escrow release reduces payment delays, eliminates manual invoicing overhead, and builds trust between shippers and carriers.

2.2 Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Non-Functional (Performance)	The system shall support a minimum of 5,000 concurrent users and respond to all search, bid-submission, and load-posting requests within 3 seconds under peak load conditions.	High	Pass: Load testing with 5,000 simulated concurrent users shows 95th-percentile response time ≤ 3 seconds for all transactional endpoints. Fail: Response time exceeds 3 seconds for more than 5% of requests under the specified load.	A freight marketplace must handle real-time bidding from thousands of carriers across the country without degradation, as delays can result in missed shipment windows.
NFR-002	Non-Functional (Security)	The system shall encrypt all payment data and personally identifiable information (PII) using AES-256 encryption at rest and TLS 1.3 encryption in transit, and shall enforce role-based access control (RBAC) for all user operations.	High	Pass: Security audit confirms AES-256 at rest, TLS 1.3 in transit, and RBAC policies correctly restrict unauthorized access across all API endpoints. Fail: Any payment data or PII is stored/transmitted in plaintext, or a user can access resources outside their role permissions.	Handling financial transactions and commercial contracts mandates strict data protection to comply with industry regulations and prevent fraud.

3. Actor Identification

SL.NO	Actor Name	Type	Category	Description
1	Shipper	Human	Primary	A commercial business entity that posts cargo loads onto the marketplace, evaluates carrier bids, awards shipment contracts, and confirms proof-of-delivery to trigger payment release.
2	Freight Carrier	Human	Primary	A trucking company or independent operator that searches for available loads, submits competitive bids, transports awarded cargo, and uploads proof-of-delivery documentation upon completion.
3	System Administrator	Human	Supporting	An internal platform operator responsible for managing user accounts, resolving disputes between shippers and carriers, monitoring system health, and configuring marketplace parameters.
4	Payment Gateway	External System	Supporting	A third-party financial service (e.g., Razorpay, Stripe) that the system interfaces with to hold escrow funds, process milestone payments, and issue transaction

				receipts to both parties.
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Note: Four actors are identified — two primary actors (Shipper, Freight Carrier) who directly interact with the core business processes, and two supporting actors (System Administrator, Payment Gateway) that enable operational and financial workflows.

4. Use-Case Summary

UC ID	Use Case Name	Primary Actor	Brief Description
UC-01	Post Cargo Load	Shipper	The shipper creates a new freight listing on the marketplace by providing cargo details (origin, destination, weight, dimensions, type, pickup date, and delivery deadline) for carriers to discover and bid on.
UC-02	Search and Bid on Load	Freight Carrier	The carrier searches the marketplace using filters (route, date, cargo type), views load details, and submits a priced bid with an estimated transit time for a selected load.
UC-03	Accept Carrier Bid	Shipper	The shipper reviews all received bids for a posted load, compares carrier profiles and ratings, and selects one winning bid to award the shipment contract.
UC-04	Upload Proof of Delivery	Freight Carrier	Upon completing the delivery, the carrier uploads a digitally signed POD document including recipient signature, timestamp, and optional photographic evidence.
UC-05	Release Escrow Payment	Payment Gateway	The system verifies the shipper's POD confirmation (or triggers auto-approval after 72 hours) and instructs the payment gateway to release the escrowed milestone funds to the carrier.
UC-06	Track Shipment	Shipper	The shipper monitors the real-time status

			and location of an in-transit shipment from pickup to delivery on an interactive dashboard.
UC-07	Resolve Delivery Dispute	System Administrator	The administrator investigates and mediates disputes raised by shippers or carriers regarding damaged goods, late delivery, or POD discrepancies.

5. UML Use-Case Diagram

The UML Use-Case Diagram is provided as a separate attachment (uml.pdf).

Diagram Relationship Summary

Relationship	From	To	Stereotype	Explanation
Include	Release Escrow Payment (UC-05)	Verify Proof of Delivery	«include»	Every payment release must first verify the uploaded POD; this step is mandatory and therefore modelled as an include.
Include	Accept Carrier Bid (UC-03)	Notify Bid Status	«include»	Accepting a bid always triggers notifications to the winning and losing carriers; this mandatory step is modelled as an include.
Extend	Resolve Delivery Dispute (UC-07)	Release Escrow Payment (UC-05)	«extend»	A dispute is raised only when delivery verification or payment release encounters a problem; this conditional behaviour extends the base escrow payment use case.

6. Detailed Use-Case Flow — Release Escrow Payment

Use Case ID	UC-05
Use Case Name	Release Escrow Payment
Goal	To securely release the milestone payment held in escrow to the Freight Carrier's bank account after verified proof-of-delivery confirmation, ensuring both parties are financially protected.
Primary Actor	Payment Gateway
Secondary Actors	Shipper, Freight Carrier
Trigger	The Freight Carrier uploads a Proof-of-Delivery (POD) document for an awarded shipment.

Preconditions

1. The shipment contract has been awarded to a carrier (UC-03 completed).
2. The shipper has funded the escrow account with the agreed bid amount.
3. The freight carrier has physically delivered the cargo to the destination.
4. The freight carrier has uploaded a valid POD document (UC-04 completed).
5. Both shipper and carrier have verified and active bank account details on file.

Postconditions

1. The milestone payment has been transferred from the escrow account to the carrier's registered bank account.
2. The shipment status is updated to "Completed — Payment Released."
3. Both shipper and carrier have received a digital payment confirmation receipt via email and in-app notification.
4. The transaction is recorded in the system's financial audit log with a unique transaction reference ID.

Main Success Scenario

Step	Actor	Action
1	Freight Carrier	Uploads the digitally signed Proof-of-Delivery document via the shipment dashboard (includes recipient name, signature image, delivery timestamp, and optional photo evidence).
2	System	Validates the POD document

		format, checks for mandatory fields (recipient name, signature, timestamp), and verifies the digital signature integrity.
3	System	Updates the shipment status from "In Transit" to "Delivered — Pending Verification" and logs the POD submission timestamp.
4	System	Sends a notification (email + in-app) to the Shipper informing them that a POD has been uploaded and is awaiting their confirmation.
5	Shipper	Logs into the platform, navigates to the shipment dashboard, and reviews the uploaded POD document — inspecting the recipient signature, delivery timestamp, and photo evidence.
6	Shipper	Confirms the POD by clicking the "Approve Delivery" button, attesting that the cargo was received in satisfactory condition.
7	System	Records the shipper's approval with a timestamp and updates the shipment status to "Delivery Confirmed."
8	System	Initiates the escrow release workflow: calculates the net payable amount (bid amount minus platform commission), generates a payment instruction, and transmits it to the Payment Gateway via a secure API call.
9	Payment Gateway	Processes the payment instruction, debits the escrow holding account, and credits the Freight Carrier's registered bank account with the net payable amount.
10	Payment Gateway	Returns a transaction confirmation response to the system, including a unique transaction reference ID, transfer amount, and settlement timestamp.
11	System	Updates the shipment status to

		"Completed — Payment Released" and records the transaction details in the financial audit log.
12	System	Sends payment confirmation receipts to both the Shipper (showing amount deducted and platform commission) and the Freight Carrier (showing net amount received and transaction reference).

Alternate Flow — Auto-Approval After 72-Hour Window

Step	Actor	Action
A1 (branches from Step 5)	System	The shipper does not log in or take any action on the POD within the 72-hour review window.
A2	System	After 72 hours have elapsed since the POD notification, the system triggers the auto-approval mechanism and logs the event as "Auto-Approved — Shipper Inaction."
A3	System	Sends a final reminder notification to the shipper informing them that the POD has been auto-approved due to the expiry of the review window.
A4	System	Proceeds to Step 8 of the Main Success Scenario to initiate the escrow release workflow.

Exception Flow — Payment Gateway Failure

Step	Actor	Action
E1	Payment Gateway	Returns an error response indicating a transaction failure (e.g., insufficient escrow balance, carrier bank account invalid, gateway timeout).
E2	System	Logs the failure with the error code, transaction details, and timestamp in the financial audit log.
E3	System	Updates the shipment status to "Payment Failed — Retry Pending" and schedules an automatic retry after 30

		minutes.
E4	System	Sends an alert notification to the System Administrator with the failure details for manual investigation if needed.
E5	System	Retries the payment instruction (up to a maximum of 3 attempts). If successful, proceeds to Step 10 of the Main Success Scenario.
E6	System	If all retry attempts fail, updates the status to "Payment Failed — Manual Intervention Required" and notifies both the shipper and carrier of the delay, along with the expected resolution timeline.